MOHAMED HESHAM KHATTAB

Mechatronics Engineer

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Transferable Iqama | Saudi Council of Engineers (SCE) Registered Engineer

Objective

Mechatronics Engineer with hands on industrial experience in the design and development of integrated mechanical and automation systems including industrial machinery and robotic applications. Skilled in mechanical design using SolidWorks and manufacturing processes, with practical experience in industrial control panels, classic control circuits, and Siemens PLC programming using TIA Portal. Capable of integrating mechanical, electrical, and control components to develop and improve automated machines and robotic systems.

Education

B.Sc. in Mechatronics Engineering | Mansoura University, Egypt **(Oct. 2020 – June. 2025)**

- **CGPA: 3.71 / 4.0** (Excellent) | Ranked 20th in Cohort.
- **Thesis:** AI-Powered Remotely Operated Underwater Vehicle for Visual Exploration & Object Retrieval.

Professional Experience

Mechatronics Engineer – New Egypt Gold **(Oct. 2025 – Sep. 2026)**

- Designed and reverse engineered machine components and spare parts using SolidWorks, applying DFM principles and supervising their fabrication for equipment restoration.
- Designed and modified industrial control panels and control circuits, developed custom PCB layouts using Proteus, and configured VFDs for industrial machine operation and control.
- Prepared CAD/CAM build files for the Flashforge WaxJet 530 wax based additive manufacturing system, supporting jewelry production and ensuring consistent print quality.
- Performed preventive maintenance, fault diagnosis, root cause analysis, and quality control to maintain the operational reliability of production equipment.

Licenses & Certifications

Saudi Council of Engineers (SCE) - Professional Accreditation **(Aug. 2026)**

- Professionally registered and accredited by the Saudi Council of Engineers as a Mechatronics Engineer.

Skills

- **Mechanical Design:** Machines Design, Mechanisms, DFM/DFA, Actuators integration, Reverse Engineering.
- **CAD & Simulation:** SolidWorks (Part, Assembly, Sheet Metal, Weldments, Drawings), Static Stress Simulation.
- **Manufacturing & Prototyping:** Additive Manufacturing, Sheet Metal Fabrication, CNC Machining.
- **Industrial Automation:** PLC Programming (TIA Portal), Classic Control, VFDs, Sensors & Actuators.
- **Microcontrollers & PCB Design**: Arduino, ESP32, Proteus, Schematic Capture, PCB Layout.
- **Soft Skills:** Teamwork, Adaptability, Fast learner, Project management, Communication, & Problem solving.

Projects

1. Remotely Operated Underwater Vehicle – Graduation Project

- Designed the structure in SolidWorks, including material selection, dimensions and thruster configuration, achieving optimal performance and stable maneuvering.
- Designed a 3-DOF robotic arm, selecting MG996R servo motors for optimal torque and precision, integrating rotary position feedback for accurate control.
- Engineered a mechanical “Sport Mode” system for adaptive thruster orientation, improving propulsion efficiency by 79% and speed by 34% during navigation and maneuvering.

2. AI-Based Autonomous Research Boat

- Designed the internal structure using metal square tubes with sheet covers, ensuring stability and durability.
- Developed a buoyancy system using sealed PVC pipes for air storage, improving overall boat flotation.
- Designed a waterproof electronics box, ensuring protection with easy access for maintenance and integration

3. Heavy Weight Combat Mobile Robot (50 kg)

- Designed the robot's weapon and power transmission, and selected suitable actuators for operation.
- Optimized the robot's exterior to minimize incoming hits while maximizing weapon reach.
- Balanced weight distribution to comply with competition requirements, enhancing stability and performance.

Publications

Design and Development of Drivetrain System for Electric Motorcycle **(Nov. 2023 – Dec. 2024)**

- Presented at 25th International Middle East Power Systems Conference (MEPCON), Published on **IEEE Xplore.**
- Performed dynamic inertia analysis and mechanical simulation using SolidWorks Motion Analysis

EXTRACURRICULAR ACTIVITIES

Momentum Team

Mechanical Design Committee Head **(Sep. 2023 – Sep. 2024)**

- Led the Mechanical Design Committee, managing recruitment, evaluations, and interviews.
- Delivered training on SolidWorks and mechanical components such as power transmission and shaft design.
- Taught key manufacturing methods, motor types, and motor selection based on load requirements.

Vice Technical Director **(Sep. 2023 – Sep. 2024)**

- Served as Deputy Technical Manager overseeing team progress in the manager's absence.
- Prepared weekly reports on each committee's performance and development.

IEEE Mansoura Student Branch

Technical Team Member in Micromuse Competition at IEEE Victoris 3.0 **(Aug. 2024 – Sep. 2024)**

- Provided continuous technical support to participating teams throughout the competition.
- Participated in judging and filtration and ensured all robots met technical requirements.

Awards & Competitions

- **1st Place** **AAKRUTI Innovation Competition 2025,** Egypt Finale. **(Oct. 2025)**
- **1st Place** **IoT-Preneur Quest competition for Graduation Projects,** IEEE Victoris 4.0. **(Sep. 2025)**
- **6th Place** **International ROV Competition (ICMCT),** Military Technical College – UMVC. **(July. 2025)**
- **4th Place** **Metal Monsters Competition,** Alamein Robotics Championship. **(Oct. 2023)**
- **3rd Place** **Autonomous Sumo Competition,** IEEE HTI. **(Sep. 2023)**
- **1st Place** **Mini Metal Monsters Competition,** Maker Faire Cairo. **(Oct. 2022)**

Language

- **Arabic:** Native.
- **English:** Very Good (Professional Working Proficiency) .